

ALVI IBN AMZAD ANIL

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Summary

Electrical and Electronic Engineering graduate from BRAC University (major: Electronics), currently a Research Assistant in the Department of Electrical and Electronic Engineering. Works at the intersection of artificial intelligence, data science, and distributed sensing systems — building machine learning and computer vision pipelines over multimodal physiological and environmental sensor data, and deploying them across edge-cloud architectures for real-time inference. Experienced across the full data workflow: sensor instrumentation and calibration, signal processing and feature engineering, model training and validation, and analysis of telemetry from distributed sensor networks. Author and co-author of six peer-reviewed Q1 journal publications spanning AI-powered cybersecurity, machine learning and digital twin simulation for energy systems, and AI/IoT-enabled distributed energy storage, and a three-time Dean's List Award recipient. Passionate about engineering education, student mentorship, and contributing to excellence in teaching, research, and academic development.

Education

Bachelor of Science in Electrical and Electronic Engineering (EEE)

BSRM School of Engineering, BRAC University

Jan 2022 – Jan 2026

- Major: Electronics
- CGPA: 3.66 / 4.00
- Dean's List Award Recipient (3 Times)

Higher Secondary Certificate (HSC)

Ispahani Public School and College, Cumilla Board

GPA: 5.00 / 5.00

2020

Secondary School Certificate (SSC)

Ispahani Public School and College, Cumilla Board

GPA: 5.00 / 5.00

2018

Professional Experience

Research Assistant (RA)

Department of Electrical and Electronic Engineering

BSRM School of Engineering, BRAC University

March 2026 – Present

- Working on a fully funded Research Seed Grant Initiative (RSGI) project on multimodal attention monitoring and stress detection using wearable sensors and computer vision.
- Building the end-to-end machine learning pipeline: synchronized multimodal data acquisition, pre-processing and artifact removal, feature engineering across physiological and visual channels, and supervised model training with held-out validation.
- Implementing edge computing for on-device inference under strict latency and power budgets, with aggregation of distributed wearable sensor streams for centralized longitudinal storage and analysis.
- Applying computer vision and digital signal processing to extract behavioural and physiological features, fusing heterogeneous modalities into a single real-time decision framework.

Research & Academic Projects

A Multimodal Framework for Attention Monitoring and Stress Detection Using Wearable Sensors and Computer Vision in Academic Setting

Research Seed Grant Initiative (RSGI) Funded Project

- Biomedical AI research project integrating EEG, PPG, and skin-temperature sensing with computer vision, machine learning, and edge computing.
- Developing a real-time multimodal framework in MATLAB and Python: data fusion across asynchronous sensor streams, feature extraction, classifier training and evaluation, and deployment to resource-constrained edge hardware.

IoT-Based Air Pollution Monitoring & Adaptive Filtration Control

Undergraduate Final Year Design Project

- Developed an IoT-based air quality monitoring system with adaptive, data-driven filtration control.
- Designed a distributed sensing architecture over a WiFi mesh network, with node-level sensor calibration, aggregation of multi-node telemetry, and automated two-stage filtration driven by analysis of the collected air-quality data.

Publications

1. The Next Generation of Energy Storage for Smart and Sustainable Power Grids: A Systematic Review

Alvi Ibn Amzad Anil, Md Reshad Al Muttaki, Sadia Afrin, Shameem Hasan

Energy Conversion and Management: X (Q1, Top 7%, IF: 8.78) | Vol. 30, Art. 101771 (2026) | **1st Author**

Focus: PRISMA-guided synthesis of 75 publications from five databases across 30 countries on AI- and IoT-enabled control of distributed energy storage; comparative assessment of machine learning and deep reinforcement learning methods (15–98% accuracy gains), IoT telemetry architectures, and hybrid storage systems, with quantitative techno-economic and simulation-to-deployment gap analysis.

DOI: <https://doi.org/10.1016/j.ecmx.2026.101771>

2. Towards Smarter Grids: A Systematic Review of Wide Area Monitoring Enhancing Stability, Protection, and Promoting Environmental Sustainability

Md Reshad Al Muttaki, Alvi Ibn Amzad Anil, Sadia Afrin, Shameem Hasan

Energy Conversion and Management: X (Q1, Top 7%, IF: 8.78) | Vol. 29, Art. 101424 (2026) | **2nd Author**

Focus: Evidence synthesis of 40 peer-reviewed studies (2019–2025) on real-time analytics over geographically distributed synchrophasor (PMU) sensor networks; AI-driven anomaly detection and transient-stability assessment (97.3% accuracy in <40 ms), latency- and noise-aware simulation-to-deployment gap analysis, and an outlook on edge and quantum computing for sub-millisecond distributed inference; MATLAB proof-of-concept simulation.

DOI: <https://doi.org/10.1016/j.ecmx.2025.101424>

3. AI-Powered Cybersecurity for Smart Grid Communication: A Systematic Review of Intrusion Detection and Threat Mitigation Systems

Sadia Afrin, Md Reshad Al Muttaki, Alvi Ibn Amzad Anil, Shameem Hasan

Energy Conversion and Management: X (Q1, Top 7%, IF: 8.78) | Vol. 29, Art. 101416 (2026) | **3rd Author**

Focus: Systematic review of AI-based intrusion detection across 40 studies; hybrid deep learning (CNN–GRU, LSTM, autoencoders) exceeding 98% detection accuracy on DDoS and false-data-injection attacks; federated learning for privacy-preserving distributed model training, edge–cloud orchestration and containerized deployment, latency/throughput trade-offs in resource-constrained edge inference, and adversarial machine learning

robustness; MATLAB 8-PMU, 300-cycle testbed.
DOI: <https://doi.org/10.1016/j.ecmx.2025.101416>

4. **State-of-Charge-Aware Charging Opportunity Detection for Electric Vehicles Using Data-Driven Learning and Digital Twin Simulation**

Md Reshad Al Muttaki, Sadia Afrin, **Alvi Ibn Amzad Anil**, Shameem Hasan

Energy Conversion and Management: X (Q1, Top 7%, IF: 8.78) | Vol. 30, Art. 101775 (2026) | **3rd Author**

Methods: SoC-COD framework combining MILP-based optimal labeling with probabilistic machine learning — a Gradient Boosting Regressor for counterfactual state-of-charge trajectories ($R^2 = 0.8297$) and an XGBoost classifier for opportunity detection (ROC-AUC = 0.9915) — validated on the California Audi EV telematics dataset under chronological holdout plus digital twin simulation; designed for on-vehicle edge inference with federated learning for privacy-preserving distributed training.

DOI: <https://doi.org/10.1016/j.ecmx.2026.101775>

5. **Demand-Based Anomaly Detection Algorithm in Electric Vehicle Charging Systems Using Machine Learning and Digital Twin Simulation**

Md Reshad Al Muttaki, Sadia Afrin, **Alvi Ibn Amzad Anil**, Shameem Hasan

Journal of Energy Storage (Q1, Top 4%, IF: 9.8) | Vol. 170, Art. 122795 (2026) | **3rd Author**

Methods: ST-DAD framework — a stacking ensemble of XGBoost, LightGBM, and Random Forest with a linear meta-learner for spatio-temporal demand forecasting, coupled to z-score-normalized residual anomaly scoring; engineered temporal, spatial, port-type, and duration features over 148,136 event-level charging sessions with leakage-free scaling and temporal holdout validation; benchmarked against five baselines (Isolation Forest, One-Class SVM, LOF, MLP, autoencoder) and a grid-aware digital twin ($F1 = 0.7906$, ROC-AUC = 0.8147).

DOI: <https://doi.org/10.1016/j.est.2026.122795>

6. **Balancing Cybersecurity, Wireless Charging, and Safety: Digital Twin-Driven Techno-Economic Analysis of Electric Vehicles**

Shameem Hasan, Abu M. Fuad, B.M. Khalid Hasan Drobo, Md Reshad Al Muttaki, Sadia Afrin, **Alvi Ibn Amzad Anil**, Omar Farrok

Energy Conversion and Management: X (Q1, Top 7%, IF: 8.78) | Vol. 31, Art. 102107 (2026) | **6th Author**

Focus: Digital twin-driven co-optimization of AI-powered intrusion detection (99.4% accuracy) against techno-economic and environmental objectives across networked vehicle-to-everything (V2X) cyber-physical infrastructure; multi-layer defense architecture analysis and simulation-based security-investment optimization (652.9% ROI at the optimal security level).

DOI: <https://doi.org/10.1016/j.ecmx.2026.102107>

Leadership & Extracurricular Activities

Secretary, Editorial and Publication Department

Robotics Club of BRAC University

May 2025 – Dec 2025

- Managed publications, communications, and outreach, creating technical and promotional content for robotics initiatives.

Slam Poetry Participant

- Participated in competitive spoken-word and creative writing events.
- Authored original poems and anonymous literary works.

Training & Certifications

Industrial Attachment Program

Neural Semiconductor, Dhaka, Bangladesh

Jan 19–23, 2026

- Successfully completed an intensive industrial attachment program focused on modern semiconductor design methodologies and integrated circuit (IC) development workflows.
- Gained practical exposure to the complete VLSI design flow, including Analog IC Design, Analog & Digital VLSI Design, IC Physical Design, RTL Synthesis, and Radio Frequency (RF) circuit design.
- Developed an understanding of semiconductor fabrication processes, front-end and back-end VLSI design, and industry-standard IC design practices used in modern semiconductor development.
- Certificate: [View Certificate](#)

Technical Skills

AI & Machine Learning	Supervised classification & regression, computer vision, feature engineering, model training & validation, multimodal data fusion
Data Science	Data analysis & visualization, signal processing, time-series and sensor-stream analytics, sensor calibration & data quality
Cloud & Distributed	Edge computing & on-device inference, distributed sensor networks, WiFi mesh networking, IoT telemetry pipelines, real-time multimodal data acquisition
Programming	Python, MATLAB
Sensors	Biosensors (EEG, PPG, skin temperature), gas sensors, sensor calibration
Electronics	Analog Electronics, Digital Electronics, Embedded Systems
Design Tools	VLSI Design, Embedded System Development, Proteus, PSpice
Research	Systematic review (PRISMA), Literature Review, Academic Writing, Technical Documentation
Engineering	IoT Systems, Smart Grid Technologies, Energy Systems
Management	Project Management, Research Coordination

Additional Information

Languages: English, Bangla

Research Interests: Machine Learning & Computer Vision, Data Science & Signal Processing, Edge & Distributed Computing, IoT & Cyber-Physical Systems, Biomedical AI, Wearable Healthcare Systems & Biosensors, Smart Grids & Energy Storage Systems, Cybersecurity for Power Systems, Embedded Systems, VLSI Design.

Hobbies: Creative Writing, Poetry, Robotics Projects, Reading Research Articles

References

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